The Ball-Proximal Method and Beyond in Hadamard Spaces

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Outline

Part 1: **Paper and Background Overview**

- Ball-Optimization Oracles
- Ball-Proximal Methods in $\mathbb{R}^n$ (Gruntkowska et al. 2025)
- Hadamard Spaces

Part 2: Selected Results from Summer Research

- Ball-Proximal Methods in Hadamard Spaces
- Regularized Methods in Hadamard Spaces (time permitting. . .)

A Different Oracle

To optimize a function f, typically assume access to an oracle

Common Examples:

- $O_0(x) = \{f(x)\}$
- $O_1(x) = \{f(x), \nabla f(x)\}$
- $O_2(x) = \{f(x), \nabla f(x), \nabla^2 f(x)\}\ldots$

which all can be computed with local information of f around x.

An oracle with a different kind of local information:

$$O_r(x) = \text{argmin}_{B_r(x)} f(x) \text{ "ball-optimization oracle"}$$

But hard to compute, often no analytical formula...

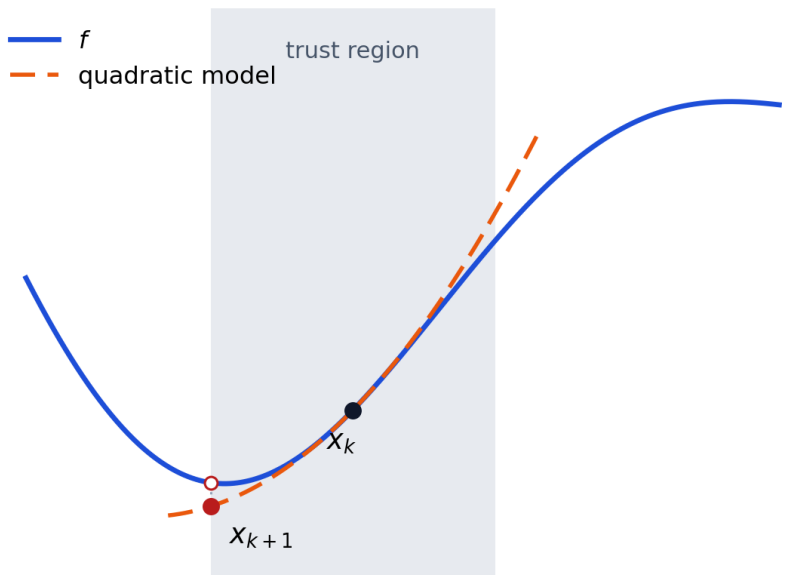
Algorithms with Ball-Optimization Oracles

Some algorithms approximate $O_r(x)$ or f.

Example: Trust-Region Methods (e.g. Conn, Gould, Toint 2000)
Approximate f with quadratic model m_k and compute

$$x_{k+1} = \operatorname{argmin}_{B_r(x)} m_k(x)$$

.



Approximate Ball-Optimization Oracles

Example: Carmon et al. 2020 design an accelerated algorithm that queries $O_r(x)$ with improved complexity

$$O(r^{-1}\log\frac{1}{\epsilon}) \to O(r^{-2/3}\log\frac{1}{\epsilon})$$

Also, Newton's Method can approximate $O_r(x)$ when Hessian is "locally stable".

Conclusion: Improvements/matching the complexity analysis of relevant problems, e.g.

- ℓ_∞ regression
- ℓ_p regression
- logistic regression

Meta-Algorithms

A meta-algorithm using $O_r(x)$: For fixed radius $r > 0$, compute

$$x_{k+1} = O_r(x_k) \equiv \operatorname{argmin}_{B_r(x_k)} f(x) \quad \text{"ball-proximal iteration"}$$

as seen in Gruntkowska et al. 2025 (but also elsewhere).

Similar in motivation to the proximal-point method:

- conceptually simple algorithm
- potential applications to practical algorithm design
- powerful algorithmic properties for convex, lsc f: linear convergence, **finite termination**. . .

Ball-Proximal Algorithm

input: initial point x_0, sequence of radii $r_k > 0$
$k = 0$
continue = true
while continue **do**
  $k = k + 1$
  choose $x_k \in \operatorname{argmin}\{f(x) : d(x, x_{k-1}) \leq r_k\}$
  if $f(x_k) = f(x_{k-1})$ **then**
    continue = false
  end if
end while
return x_k

Ball-Proximal Algorithm: Bookkeeping

Check the following for f convex, lsc:

- existence of x_k
- stopping condition
- convergence

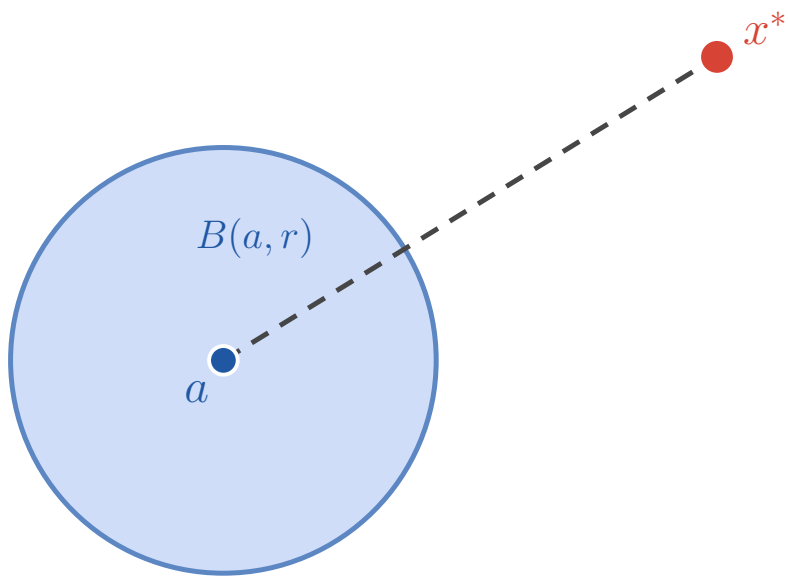
First pass: the machinery in Gruntkowska et al., 2025 in $\mathbb{R}$.

Existence and Stopping Condition

Proposition

The value $\inf_{B_r(a)} f$ *is attained, and equals* $f(a)$ *if and only if* $a \in X^*$.

Proof.



□

"Brox" Inequality

A key result used by Gruntkowska et al., 2025:

Proposition

Suppose $u \in O_r(x)$. Then there exists $c_r(x) \geq 0$ such that

- $c_r(x)(x-u) \in \partial f(u)$
- $f(y) - f(u) \geq c_r(x)\langle x-u, y-u\rangle, \forall y \in \mathbb{R}^d$

Proof.

Optimality condition + subgradients...

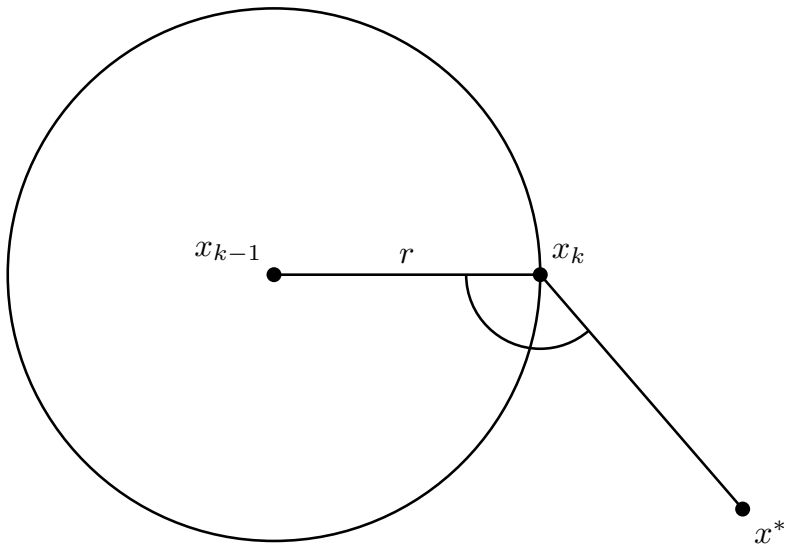
Angle Condition

Proposition

Suppose $x_k \notin X^*$. For any $x^* \in X^*$, $\angle x_{k-1} x_k x^*$ is at least $\frac{\pi}{2}$, or

$$\|x_k - x^*\|^2 \leq \|x_{k-1} - x^*\|^2 - r_k^2$$

so x_k is Fejér monotone.



Proof.

"Brox" inequality...

Finite Termination

Remark: The telescoped inequality

$$\|x_k - x^*\|^2 \leq \|x_0 - x^*\|^2 - \sum_{i=1}^{k-1} r_k^2$$

implies finite termination when $r_k = r > 0$.

Functional Decrease

Proposition

For $k \geq 1$, if $x_k \notin X^*$,

$$f(x_k) - f^* \leq \left(1 - \frac{r_k}{\|x_{k-1} - x^*\|}\right)(f(x_{k-1}) - f^*)$$

Proof.

"Brox" inequality...

Remark: Since x_k is Fejér monotone, telescoping this implies

$$f(x_k) - f^* \leq \exp(-\sum_{i=1}^{k}(\frac{r_i}{\|x_0 - x^*\|}))(f(x_0) - f^*)$$

$\sum r_k = \infty$ implies $f(x_k) \to f^*$. If $r_k = r > 0$, $f(x_k)$ converges linearly.

Hadamard Spaces

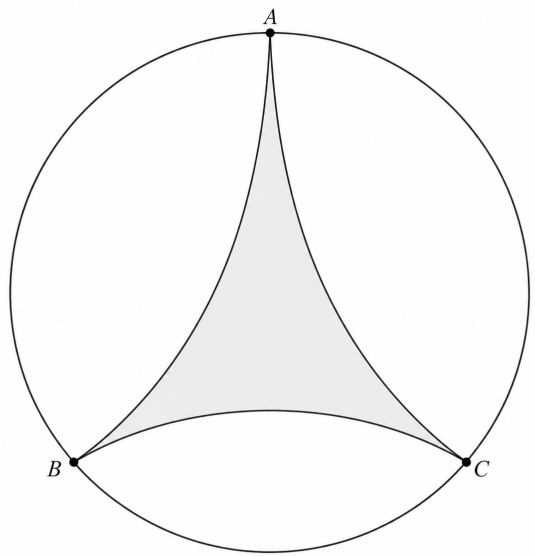
A geodesic $\gamma : [0,1] \to X$ satisfies
$d(\gamma(s), \gamma(t)) = |s - t|\, d(\gamma(0), \gamma(1))$.

CAT(0) property holds if for any geodesic γ, $y \in X$

$$t \to d^2(\gamma(t), y) \text{ is strongly convex}$$

i.e. triangles are "skinny"

A CAT(0) complete metric space is a Hadamard space.



Examples and Applications of Hadamard Spaces

Example: $\mathbb{R}^n$

Example: Hyperbolic space: unit ball with the distance $d(x,y) = \cosh^{-1}\left(1 + 2\frac{\|x-y\|^2}{(1-\|x\|^2)(1-\|y\|^2)}\right)$

- Text-embedding (Nickel and Kiela 2017)

Example: Tree Spaces (not a manifold)

- Phylogenetics (Billera, Holmes, and Vogtmann 2001)

Remark: Hadamard spaces lack dual constructions, requiring different proofs

- Babu et al. 2026, Gruntkowska et al. 2025, use inner products machinery

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- Regularized Methods in Hadamard Spaces (time permitting. . .)

Angle Condition in Hadamard Spaces

Proposition

Suppose $x_k \notin X^*$. For any $x^* \in X^*$, $\angle x_{k-1} x_{kk} x^*$ is at least $\frac{\pi}{2}$, or

$$d(x_k, x^*)^2 \leq d(x_{k-1}, x^*)^2 - r_k^2$$

Proof.

Observe that

$$r_k^2 = d^2(x_k, x_{k+1}) \leq d^2(x_k, (1-t)x_{k+1} + tx^*)$$

and apply strong convexity. . . □

Complexity

Proposition

If the convex function f has a nonempty set of minimizers X^*, then the algorithm with $r_k = r$ returns a minimizer after at most

$$1 + \left\lceil \frac{\operatorname{dist}^2(x_0, X^*)}{r^2} \right\rceil$$

iterations.

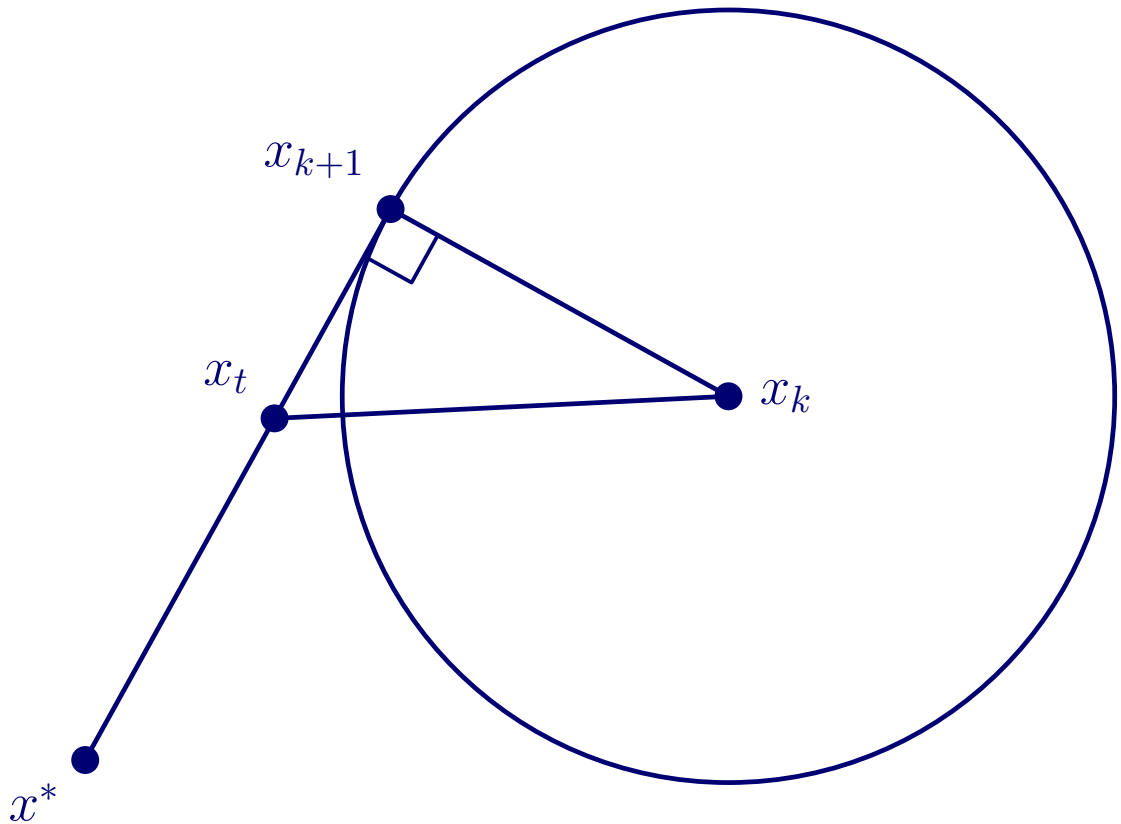
Does there exist a function which achieves this complexity?

Orthogonality Condition

Proposition

If $d^2(x_k, x^*) - d^2(x_{k+1}, x^*) = r_k^2$, *then* $x_{k+1} \in X^*$.

Proof.



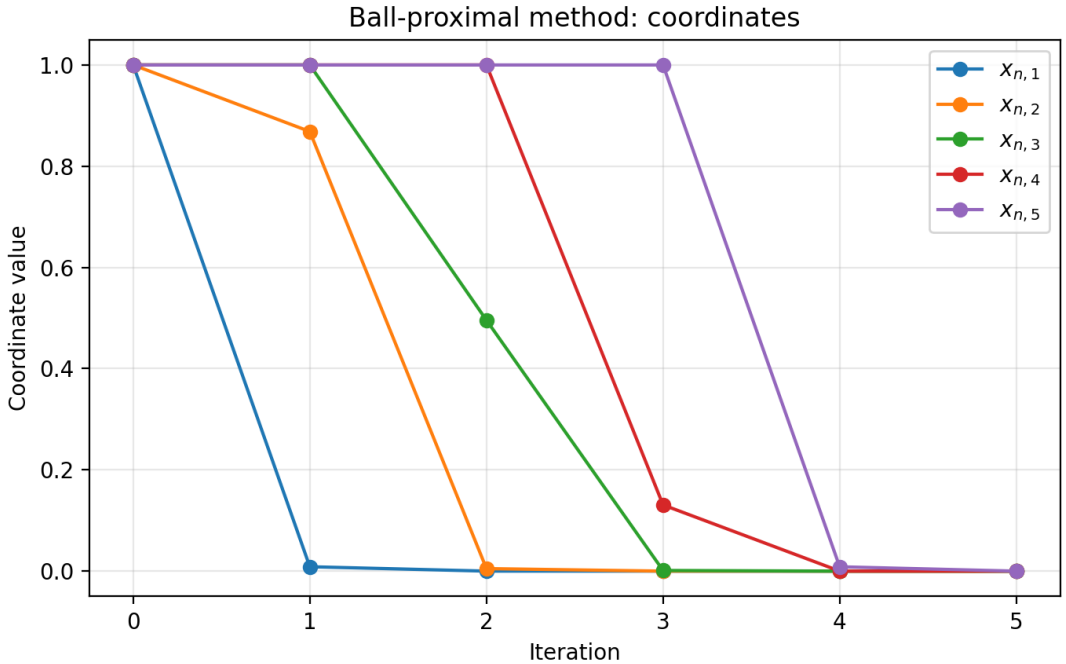
Approximate Bound

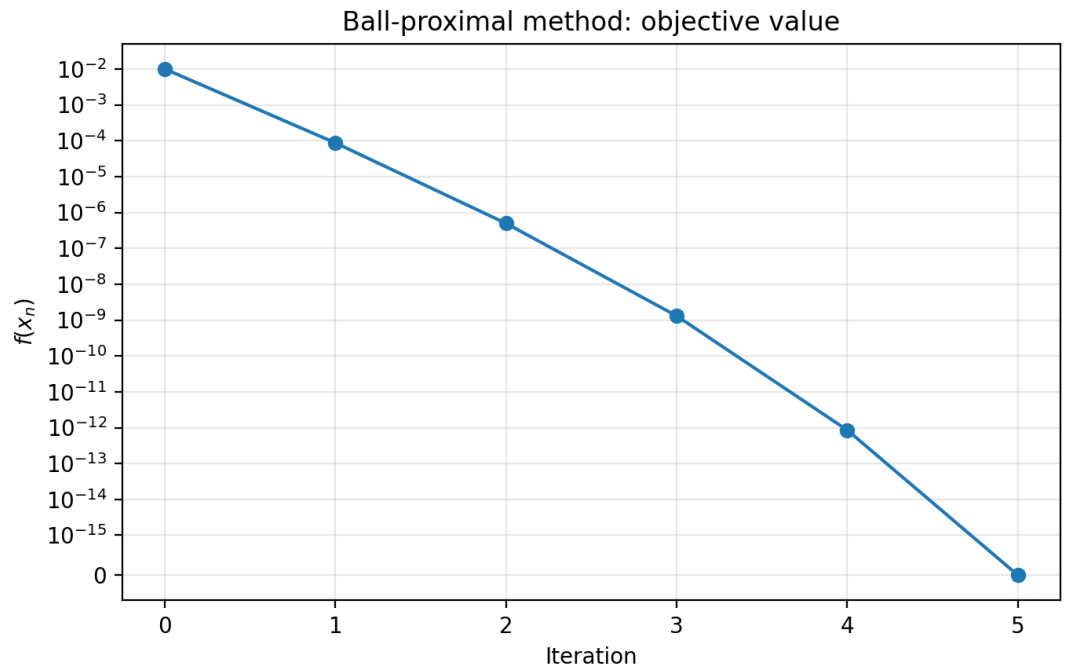
Let $X = \mathbb{R}^k$ with the max-of-linear function

$$f(x) = \max\{|c_i x_i|\}$$

with initial point $x_0 = (r, \ldots, r)$.
Complexity result guarantees $k + 1$ iterations to reach minimizer.

Informal Argument:
Select c_i to be exponentially decaying. At each iterate the algorithm propagates through at 2 coordinates, a nearly orthogonal step...


Ball-proximal method: coordinates
Coordinate value
Iteration
$x_{n,1}$
$x_{n,2}$
$x_{n,3}$
$x_{n,4}$
$x_{n,5}$


Ball-proximal method: objective value
$f(x_n)$
Iteration

Functional Residual Decrease

Proposition (Functional Residual)

For $k \geq 1$, if $x_k \notin X^*$

$$f(x_k) - f^* \leq (1 - \frac{r_k}{\operatorname{dist}(x_0, X^*)})(f(x_{k-1}) - f^*)$$

Proof.

Select t

$$f(x_{k+1}) - f^* \leq f(tx_k + (1-t)x^*) - f^* \leq t(f(x_k) - f*) \ldots$$

□

As before, $\sum r_k = \infty$ implies $f(x_k) \to f^*$. If $r_k = r > 0$, $f(x_k)$ converges linearly.

Combined Regularized Objective

Consider the minimization of the regularized objective

$$\inf[f(x) + \frac{1}{c}h(cd(a,x))].$$

where $h : \mathbb{R}^+ \to \mathbb{R}^+ \cup \{\infty\}$ is

- convex, lsc, proper
- monotone
- $h(0) = 0$
- $h'^+(0) = \lim_{t\downarrow 0} \frac{h(t)}{t} = 0$
- $\lim_{t\to\infty} \frac{h(t)}{t} = \infty$

Remark: As $c \to 0$, $\frac{1}{c}h(cd(a,x)) \to 0$, and when $c \to \infty$, then $\frac{1}{c}h(cd(a,x)) \to \infty$.

Examples of Regularizers

Generalizes standard algorithms:

- $h(x) = x^2$: proximal point method
- $h(x) = \mathbf{1}\{x \leq 1\}$: ball-proximal method
- $h(x) = x^p$: regularizers with degree $p > 1$

Can define an algorithm which generalizes both by computing

$$x_k \in \operatorname{argmin}\{f(x) + \frac{1}{c_k} h(c_k d(x, x_{k-1}))\}$$

and recover similar convergence results which adapt to h...

Angle Condition

Proposition

Assume $x_{k+1} \notin X^*$. For any $x^* \in X^*$, $\angle x_k x_{k+1} x^*$ is at least $\frac{\pi}{2}$, or

$$d(x_{k+1}, x^*)^2 \leq d(x_k, x^*)^2 - d^2(x_k, x_{k+1})$$

Proof.

By optimality and convexity
$h(c_{k+1} d(x_k, x_{k+1})) < h(c_{k+1} d(x_k, (1-t)x_{k+1} + tx^*))$, implying
$d^2(x_k, x_{k+1}) \leq d^2(x_k, (1-t)x_{k+1} + tx^*) \ldots$ □

Functional Residual

Proposition

The algorithm has the following rate of decrease:

$$\frac{1}{c_{k+1}} h^*\left(\frac{F_{k+1}}{d(x^*, x_0)}\right) \leq F_k - F_{k+1}$$

Proof.

By convexity and the optimality of x_{k+1},

$$f(x_{k+1}) + \frac{1}{c_{k+1}} h(cd(x_k, x_{k+1})) \leq (1-t) f(x_k) + t f^* + \frac{1}{c_{k+1}} h(tcd(x^*, x_k))$$

Rearrange and maximize the left-hand side...

□

Appendix

Weak Convergence for Ball-Proximal Method

There is also an analogous result for convergence of the iterates in Hadamard spaces when r_k grows sufficiently fast.

A sequence x_n weakly converges to x if for every geodesic γ with $x \in \gamma$, the projection $P_\gamma x_n \to x$ as $n \to \infty$.

Proposition

If $\sum r_k = \infty$, then x_n weakly converges to a minimizer.

Proof.

x_n is a minimizing sequence that is also Fejér monotone. Then use a standard argument... □

Weak Convergence for General Regularizer

Proposition

If $\sum \frac{1}{c_k} = \infty$, *then* x_n *weakly converges to a minimizer.*

Proof.

Telescope the functional residual inequality to get

$$h^*\left(\frac{F_\infty}{d(x^*, x_0)}\right) \sum_{i=1}^{k} \frac{1}{c_i} \leq \sum_{i=1}^{k} \frac{1}{c_i} h^*\left(\frac{F_i}{d(x^*, x_0)}\right) \leq F_0 - F_k$$

and use properties of h^*... □